THE DIRECT-DECORATION ROOTED-END COMPACTIFICATION VERSION 1.1 VISUAL EDITION

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Abstract. Let f be a quadratic polynomial at a Kahn–Lyubich direct-decoration scale. Under an explicit finite-depth graph-avoidance hypothesis, the construction provides a Jordan puzzle piece P, two disjoint univalent pullbacks $Q_0, Q_1 \subset P$ of P under $g = f^q$, and a primitive renormalization continuum K_ϵ in the complementary gap. Iterating the inverse branches produces finite fibers K_w and nested end fibers E_ω. We prove that their union $\mathcal{H}_\partial$ is compact, that these fibers are precisely its connected components, and that the component quotient is the binary end compactification. Away from the root, g is a two-sheeted covering semiconjugate to the one-sided shift. Every component is full and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected. We construct disjoint Jordan cells in which fibers at an arbitrary set of finite addresses may be replaced while preserving the component quotient and, for full replacements, connected complement. Finally, finitely many protected prime ends can be transported to a full carrier meeting $\mathcal{H}_\partial$ in precisely the selected fibers. No renormalization structure on the end fibers is asserted.

1. Introduction

The MLC conjecture—local connectivity of the Mandelbrot set—has been reduced, over three decades, to renormalization statements of increasingly combinatorial character [DH85, Hub93, Lyu97, Kah06, KL08, KL09, KKL26]. The present frontier is unbounded satellite combinatorics: traditional complex a priori bounds cannot be uniform along satellite relative periods tending to infinity [BLOT22], and the current proposals replace bounds by structural transition statements for towers of renormalizations. In Dudko's program [Dud25], the central objects are *virtual Julia sets*: connected hulls of Cantor families of small Julia sets carrying a finite part of the critical orbit. Kahn–Kapiamba–Lyubich [KKL26] construct, in their native primitive category, a Hubbard tree inside the full dynamical carrier, lifted from the locally connected rational lamination model when the carrier itself is not locally connected.

The present paper studies the *first direct-decoration scale* of Kahn–Lyubich [KL08, KL09]: a quadratic polynomial f chosen in a non-principal decoration outside a directly attached satellite copy of the Mandelbrot set, whose modified principal nest terminates in a primitive quadratic-like renormalization. There the puzzle supplies a separator piece P, an iterate $g = f^q$, and two disjoint closed univalent pullbacks $Q_0, Q_1 \subset P$ under the finite-depth graph-avoidance input (2.5); the nest supplies a terminal primitive root continuum K_ϵ lying in the first-level gap $P \setminus (Q_0 \cup Q_1)$. This produces a concrete planar model for the component and access data that arise in virtual-renormalization programs.

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1.1. **Main theorem and scope.** Write $\phi_a\colon P \to Q_a$ $(a \in \{0,1\})$ for the inverse branches of g, $\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}$ for a finite binary word $w = a_1 \cdots a_m$, and

$$Q_w = \phi_w(P), \qquad K_w = \phi_w(K_\epsilon), \qquad E_\omega = \bigcap_{m\geq 1} Q_{\omega|m} \quad (\omega \in \{0,1\}^{\mathbb{N}}).$$

The binary address compactification Ω_∂ consists of all finite words and all infinite words, topologized as a compact subset of $\{0,1,\#\}^{\mathbb{N}}$. The *rooted-end compactum* is

$$\mathcal{H}_\partial = \bigcup_w K_w \;\cup\; \bigcup_\omega E_\omega.$$

Under Standing Hypotheses 2.2, $\mathcal{H}_\partial$ is a compact subset of $K(f)$ whose components are precisely the fibers K_w and E_ω (Theorem 4.3). The component projection $\pi_\partial\colon \mathcal{H}_\partial \to \Omega_\partial$ is continuous and closed, and $g\colon \mathcal{H}_\partial \setminus K_\epsilon \to \mathcal{H}_\partial$ is a two-sheeted covering descending to first-symbol deletion on Ω_∂ (Theorems 4.4 and 4.5). Geometric nesting instead deletes the last symbol; Section 4.5 records the distinction and the separate transition at the root.

Every fiber is full, and both $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ and the relevant relative complements are connected. Pairwise disjoint Jordan cells allow fibers at an arbitrary set of finite addresses to be replaced by connected continua without changing the component quotient; full replacements preserve connected complement (Theorems 5.1, 5.4 and 5.5). For a finite set of selected fibers, protected prime ends transport to a full carrier H satisfying

$$H \cap \mathcal{H}_\partial = \bigcup_{w\in F} K_w$$

and retaining the cyclic order of the protected chains (Theorem 6.8).

The end compactification of a binary tree is classical. The point here is its realization by inverse branches, puzzle pullbacks, and nested puzzle intersections in one dynamical plane. The result is a one-scale model, not a construction of Dudko's virtual Cantor family: end fibers are not identified with small Julia sets, and no self-return map on an individual end fiber is constructed. The construction is finite only at the connector stage; the replacement theorem itself permits infinitely many addresses.

Section 2 extracts the direct-decoration data. Sections 3–5 construct the compactum and prove its dynamical and replacement properties. Section 6 constructs protected carriers, and Section 7 records the limitations and open direction.

2. Dynamical setup and realization

2.1. **Dynamical setting.** Let $f = f_c(z) = z^2 + c$ with c in a non-principal decoration attached to a directly attached satellite copy $\mathcal{M}_{t/q}$, $q > 1$, of the Mandelbrot set, with c outside the copy itself. In the Kahn–Lyubich decoration puzzle [KL08], let Y^1 denote the critical depth-one piece and Z^1_j the off-critical depth-one pieces, and set

$$n = \min\{m \geq 1 : f^{mq}(0) \in \operatorname{int} Z^1_j \text{ for some } j\}. \tag{2.1}$$

The minimum is finite because c lies outside the satellite copy. The terminal component Z^1_κ and the intermediate pullback components determine the escape route, and the selected escape piece is the critical piece

$$V_{\mathrm{esc}} = f^{-nq}\big(Z^1_\kappa\big)\big|_0 = Y^{nq+1}. \tag{2.2}$$

Kahn–Lyubich define this piece as the *first kingdom* and prove that every domain of its first-return map is compactly contained in it [KL08, KL09].

We assume that these puzzle and dynasty data are defined and that the modified principal nest built on (2.2) terminates in the stated primitive quadratic-like renormalization. All subsequent abstract results use only Standing Hypotheses 2.2, except for the explicitly labeled Julia-set upgrades under (H3). The terminal datum is a polynomial-like first-return branch

$$f^p\colon A_0 \longrightarrow B, \qquad A_0 \Subset B, \tag{2.3}$$

of some period p, where A_0 is the critical first-return domain containing 0 [KL09, DH85]. Let K_ϵ denote its filled Julia set (the *terminal primitive root*) and let $C_i = f^i(K_\epsilon)$, $0 \le i < p$, denote the terminal cycle, so $C_0 = K_\epsilon$.

Proposition 2.1 (Critical-base cycle exclusion)**.** *For $1 \le i < p$ we have $C_i \cap B = \varnothing$. The complete cycle itinerary is*

$$C_0 \longrightarrow C_1 \longrightarrow C_2 \longrightarrow \cdots \longrightarrow C_{p-1} \longrightarrow C_0.$$

Proof. Equation (2.3) is a first-return branch, so $f^i(A_0) \cap B = \varnothing$ for $1 \le i < p$. Since $K_\epsilon \subset A_0$, the same holds for $C_i = f^i(K_\epsilon)$. The itinerary follows from the definition of C_i and the complete invariance of the filled Julia set of the branch $f^p\colon A_0 \to B$. □

2.2. **The data and hypotheses.** The compactification uses the following data extracted from the construction above. Let

$$P = Y^{(n-1)q+1}, \qquad g = f^q, \qquad g\colon Q_a \longrightarrow P \quad (a \in \mathtt{A} := \{0,1\}) \tag{2.4}$$

where Q_0, Q_1 are the two univalent pullbacks of P under g inside P supplied by the decoration construction [KL08]: each Q_a is the closure of a component of $g^{-1}(\operatorname{int} P)$, and the two components are distinct. Let $V^0 = Y^{nq+1}$ be the first kingdom (2.2), so that $K_\epsilon \Subset \operatorname{int} V^0$ and V^0, Q_0, Q_1 are distinct puzzle pieces of the common depth $nq+1$. Write $\phi_a\colon P \to Q_a$ for the inverse branches.

For the q-level realization we impose one additional hypothesis. Let Γ^d be the depth-d puzzle graph and assume

$$\{f^j(0) : j \ge 0\} \cap \Gamma^d = \varnothing \qquad (0 \le d \le nq+1). \tag{2.5}$$

Thus the critical orbit avoids every puzzle graph through the deepest level used here. The deeper pullbacks in Remark 2.5 remove this hypothesis at the cost of changing the transition time from q to nq.

Standing Hypotheses 2.2. Throughout the paper, (H1), (H2), and (H4) hold:

(H1) P, Q_0, Q_1 are *closed* Jordan disks with $Q_0 \cup Q_1 \subset P$, and the two closed pieces are disjoint: $Q_0 \cap Q_1 = \varnothing$. Disjointness of the closed pieces, not merely of their interiors, is used.

(H2) Each restriction $g|_{Q_a}$ is proper of degree one and extends to a homeomorphism $g\colon Q_a \to P$ of the closed Jordan pieces; equivalently, each $\phi_a\colon P \to Q_a$ is an orientation-preserving homeomorphism of closed Jordan disks.

(H4) The root K_ϵ is a nondegenerate full continuum (a full continuum containing more than one point) with

$$K_\epsilon \subset \operatorname{int} P \setminus (Q_0 \cup Q_1); \tag{2.6}$$

it lies in the *first-level gap*. We write G_0 for the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing K_ϵ.

Optional Julia hypothesis. When explicitly invoked, we additionally assume:

(H3) $K(f) = J(f)$: the filled Julia set of f has empty interior.

Standing scope. The following limitations remain in force:

(N1) The root transition $\tau_{\mathrm{root}}\colon K_\epsilon \mapsto f^q(K_\epsilon)$ is separate data (Section 4.5); it is *not* an extension of the first-symbol deletion map σ_{pre}, and no assertion of permanent escape from $\mathcal{H}_\partial$ is made.
(N2) No second satellite portrait is constructed, and the end fibers E_ω are *not* identified with small Julia sets.
(N3) No analytic return map *on* an end fiber is constructed: no first-return or polynomial-like self-return structure on any single E_ω is claimed. The analytic transitions $E_{a\omega} \to E_\omega$ of (4.5) *are* proved; they are a different assertion.
(N4) No uniform (period-independent) incidence or marking bound is claimed.

2.3. **Realization of the hypotheses.** Closed-piece disjointness in (H1) is not formal vocabulary: closed puzzle pullbacks may touch at boundary points [KL08]. The following local lemma supplies the needed exclusion.

Lemma 2.3 (Local uniqueness of a regular closed pullback)**.** *Let h be a holomorphic map defined near $z \in \mathbb{C}$ with $h'(z) \neq 0$, and let P be a closed Jordan disk containing $h(z)$. Then z belongs to the closure of at most one component of $h^{-1}(\operatorname{int} P)$.*

Proof. Choose an open topological disk V about $h(z)$ and a neighborhood W of z such that $h\colon W \to V$ is a homeomorphism. If $h(z) \in \operatorname{int} P$, shrink V so that $V \subset \operatorname{int} P$; then $V \cap \operatorname{int} P = V$ is connected. If $h(z) \in \partial P$, apply the Schoenflies theorem to a chart in which ∂P near $h(z)$ is a straight segment and shrink V to a round disk in that chart; then $V \cap \operatorname{int} P$ is an open half-disk, again connected. In both cases

$$W \cap h^{-1}(\operatorname{int} P) = (h|_W)^{-1}\big(V \cap \operatorname{int} P\big)$$

is connected. If a component C of $h^{-1}(\operatorname{int} P)$ has $z \in \overline{C}$, then C meets the connected set $W \cap h^{-1}(\operatorname{int} P)$, which is therefore contained in C. Two distinct components would both contain this nonempty connected set, contradicting disjointness of components. □

Lemma 2.4 (Realization under graph avoidance)**.** *Under (2.5), the decoration data (2.4) satisfy (H1), (H2), and (H4).*

Proof. The pieces P, Q_0, Q_1 are closed Jordan puzzle pieces and the two pullbacks are closures of distinct components of $g^{-1}(\operatorname{int} P)$ [KL08]. Suppose the closed pieces Q_0, Q_1 met at a point z. Since the interiors of distinct depth-$(nq+1)$ pieces are disjoint complementary components of the depth-$(nq+1)$ puzzle graph, z lies on that graph. We claim z is a regular point of $g = f^q$. Indeed, the critical points of g are the points ζ with $f^i(\zeta) = 0$ for some $0 \le i < q$, and f maps the depth-d puzzle graph into the depth-$(d-1)$ graph; if z were critical, $0 = f^i(z)$ would therefore lie on a puzzle graph. But (2.5) excludes that possibility. Hence z is a regular point of g, and Lemma 2.3 puts z in the closure of at most one component of $g^{-1}(\operatorname{int} P)$. Since $z \in Q_0 \cap Q_1$ and the Q_a are closures of distinct components, this is a contradiction. This proves (H1).

A proper degree-one holomorphic map between Jordan domains is a conformal isomorphism of the interiors; since the pieces are closed Jordan disks bounded by finitely many ray and equipotential arcs, Carathéodory extension gives a homeomorphism of the closed pieces, orientation-preserving because the interior map is holomorphic. This proves (H2).

For (H4): the filled Julia set of the polynomial-like restriction (2.3) is a full continuum [DH85]. Nondegeneracy is automatic: K_ϵ is the filled Julia set of a degree-two polynomial-like branch, whose Julia set is a perfect set [DH85], so K_ϵ is not a point. The three pieces V^0, Q_0, Q_1 are distinct puzzle pieces of the common depth $nq+1$; the interior of each is a complementary component of the depth-$(nq+1)$ puzzle graph, so $\operatorname{int} V^0$ is disjoint from the

closed pieces Q_0 and Q_1. Since $K_\epsilon \Subset \operatorname{int} V^0$ and $V^0 \subset P$ (deeper critical pieces are nested), we get (2.6). □

Remark 2.5 (Deeper pullbacks). Kahn–Lyubich construct, for every vertex v of P, a univalent f^{nq}-pullback $Q^v \subset P$ and state that this deeper family is *pairwise disjoint* [KL08, Lemma 2.1]. These pieces are components of the domain of $(f^q|_{Q_0\cup Q_1})^n$, hence lie in $Q_0 \cup Q_1$ and remain disjoint from $K_\epsilon \Subset V^0$. Choosing two Q^v and setting $g = f^{nq}$ therefore preserves (H1), (H2), and (H4). All subsequent results then hold with transition time nq in place of q, and the root arrow becomes $f^{nq}(K_\epsilon) = C_{nq \bmod p}$.

3. Compactification tools

For a compact set $K \subset \mathbb{C}$ let

$$\operatorname{Fill}(K) = K \cup \bigcup\{\text{bounded components of } \mathbb{C} \setminus K\}. \tag{3.1}$$

A compact connected set $K \subsetneq S^2$ is *full* if $S^2 \setminus K$ is connected. For bounded $K \subset \mathbb{C}$, viewing $\mathbb{C} \subset \widehat{\mathbb{C}} = S^2$, this is equivalent to $\operatorname{Fill}(K) = K$.

The first lemma is the Čech form of Alexander duality. Hatcher explains the passage from the locally contractible case to arbitrary compacta [Hat02, pp. 256–257]; the underlying Čech theory is developed in Eilenberg–Steenrod [ES52]. For every nonempty proper compact $K \subset S^2$,

$$\widetilde{H}_0(S^2 \setminus K; \mathbb{Z}) \;\cong\; \check{H}^1(K; \mathbb{Z}). \tag{3.2}$$

The Čech form, rather than the locally contractible special case, is required: the compacta below are Julia-set pullbacks with no local structure assumptions.

Lemma 3.1 (Finite disjoint full continua)**.** *If $K_0, \dots, K_N \subset S^2$ are pairwise disjoint nonempty full continua, then $S^2 \setminus \bigcup_{r=0}^{N} K_r$ is connected.*

Proof. Fullness of K_r makes $S^2 \setminus K_r$ connected, so (3.2) gives $\check{H}^1(K_r; \mathbb{Z}) = 0$. The finitely many pairwise disjoint compacta have pairwise disjoint open neighborhoods; the covers of $\bigcup_r K_r$ subordinate to these neighborhoods are cofinal among all open covers, and each has nerve equal to the disjoint union of the individual nerves. Passing to first cohomology and then to the direct limit defining Čech cohomology,

$$\check{H}^1\Big(\bigcup_r K_r; \mathbb{Z}\Big) \cong \bigoplus_r \check{H}^1(K_r; \mathbb{Z}) = 0.$$

A second application of (3.2) makes the complement connected. □

Lemma 3.2 (Nested finite unions of full continua)**.** *Let $F_0 \supset F_1 \supset \cdots$ be compact proper subsets of S^2, each a finite union of pairwise disjoint full continua. Then $\bigcap_m F_m$ has connected complement in S^2. If moreover U is a Jordan domain with $F_m \Subset U$ for every m and each $F_m \cup (\widehat{\mathbb{C}} \setminus U)$ is again a finite union of pairwise disjoint full continua, then $U \setminus \bigcap_m F_m$ is connected.*

Proof. By Lemma 3.1 every $S^2 \setminus F_m$ is connected and nonempty. These complements increase with m, all contain a common point (any point of $S^2 \setminus F_0$), and their union is the complement of the intersection; an increasing union of connected sets with a common point is connected. For the localized statement, apply the same argument to the compacta $F_m \cup (\widehat{\mathbb{C}} \setminus U)$: the complement of the intersection is then exactly $U \setminus \bigcap_m F_m$. □

Lemma 3.3 (Pointed Schoenflies transport)**.** *Let $\phi\colon D \to D'$ be an orientation-preserving homeomorphism of closed Jordan disks in $\mathbb{C}$. Then ϕ extends to an orientation-preserving homeomorphism Φ of $S^2 = \widehat{\mathbb{C}}$ with $\Phi(\infty) = \infty$. Consequently, for every compact $S \subset D$:*

(a) *if S is a full continuum, so is $\phi(S)$; and*
(b) $\Phi(\mathrm{Fill}(S)) = \mathrm{Fill}(\phi(S))$*; in particular* $\mathrm{Fill}(\phi(S)) = \phi(\mathrm{Fill}(S))$ *whenever* $\mathrm{Fill}(S) \subset D$.

Proof. The complements $\Delta = S^2 \setminus \operatorname{int} D$ and $\Delta' = S^2 \setminus \operatorname{int} D'$ are closed Jordan disks in S^2 containing ∞ in their interiors. By the Schoenflies theorem choose homeomorphisms $h\colon \Delta \to \overline{\mathbb{D}}$ and $h'\colon \Delta' \to \overline{\mathbb{D}}$; composing with self-homeomorphisms of $\overline{\mathbb{D}}$ that fix the boundary pointwise, arrange $h(\infty) = h'(\infty) = 0$. The circle homeomorphism $\beta = h' \circ (\phi|_{\partial D}) \circ h^{-1}$ extends by coning, $\widehat{\beta}(re^{i\theta}) = r\,\beta(e^{i\theta})$, to a homeomorphism of $\overline{\mathbb{D}}$ fixing 0. Define $\Phi = \phi$ on D and $\Phi = (h')^{-1} \circ \widehat{\beta} \circ h$ on Δ; the two definitions agree on ∂D, so Φ is a sphere homeomorphism, and $\Phi(\infty) = (h')^{-1}(0) = \infty$. It is orientation-preserving because it is so on the open disk $\operatorname{int} D$.

For (b): Φ carries the components of $\widehat{\mathbb{C}} \setminus S$ bijectively onto the components of $\widehat{\mathbb{C}} \setminus \phi(S)$ and matches the component of ∞ with the component of ∞; hence it carries the union of S with its bounded complementary components onto the corresponding union for $\phi(S)$. When $\mathrm{Fill}(S) \subset D$, the restriction of Φ to $\mathrm{Fill}(S)$ is ϕ itself, giving the displayed identity with ϕ. (a) is the special case in which S has no bounded complementary components: then neither does $\phi(S)$, and $S^2 \setminus \phi(S) = \Phi(S^2 \setminus S)$ is connected. □

Lemma 3.4 (Full-continuum Jordan neighborhood)**.** *If a nondegenerate full continuum K is compactly contained in an open set $G \subset \mathbb{C}$, there is a closed Jordan disk D with*

$$K \subset \operatorname{int} D \subset D \Subset G. \tag{3.3}$$

Proof. Uniformize the simply connected exterior by $\psi\colon \{|z| > 1\} \to \widehat{\mathbb{C}} \setminus K$. The filled interiors of the analytic Jordan curves $\psi(\{|z| = R\})$, $R > 1$, are closed Jordan disks decreasing to K as $R \downarrow 1$. If none lay in G, compactness would place a point of the closed set $\widehat{\mathbb{C}} \setminus G$ in their intersection K, a contradiction. □

4. The rooted-end compactification

4.1. **Addresses and fibers.** Let $\mathtt{A} = \{0, 1\}$, let $\mathtt{A}^{<\mathbb{N}}$ denote the finite binary words including the empty word ϵ, and let $\mathtt{A}^{\mathbb{N}} = \mathtt{A}^{\mathbb{N}}$ denote the ends. For $w = a_1 \cdots a_m \in \mathtt{A}^{<\mathbb{N}}$ put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \qquad Q_w = \phi_w(P), \qquad K_w = \phi_w(K_\epsilon), \tag{4.1}$$

with $\phi_\epsilon = \mathrm{id}$, $Q_\epsilon = P$, $K_\epsilon = K_\epsilon$. For $\omega \in \mathtt{A}^{\mathbb{N}}$ put

$$E_\omega = \bigcap_{m \geq 1} Q_{\omega|m}. \tag{4.2}$$

The *binary address compactification* is

$$\Omega_\partial = \{w\#^\infty : w \in \mathtt{A}^{<\mathbb{N}}\} \cup \mathtt{A}^{\mathbb{N}} \;\subset\; \{0, 1, \#\}^{\mathbb{N}}, \tag{4.3}$$

with the product topology; we identify w with $w\#^\infty$. The *rooted-end compactum* is

$$\mathcal{H}_\partial = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} K_w \;\cup\; \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega. \tag{4.4}$$

Write $\mathrm{fib}(w) = K_w$ and $\mathrm{fib}(\omega) = E_\omega$ for the realization map on Ω_∂.

Two words are *comparable* if one is a prefix of the other (written $u \sqsubseteq v$), and an end ω *extends* u if $\omega|_{|u|} = u$.

Lemma 4.1 (Cylinder separations)**.** *Under Standing Hypotheses 2.2:*

(a) $Q_{wa} = \phi_w(Q_a) \subset Q_w$; *appending symbols descends through nested closed Jordan disks, and* $g^{|w|}\colon Q_w \to P$ *is a homeomorphism.*
(b) *If* u, v *are incomparable, then* $Q_u \cap Q_v = \varnothing$.
(c) *If* u *is a strict prefix of* v*, then* $K_u \cap Q_v = \varnothing$. *In particular* K_u *is disjoint from* K_v *and from every* E_ω *with* ω *extending* u.
(d) *Distinct fibers of the realization map are pairwise disjoint.*

Proof. (a) is immediate from (H2) and $\phi_{wa} = \phi_w \circ \phi_a$. For (b), write $u = cas$, $v = ca't$ with $a \neq a'$ for the longest common prefix c; then $Q_u \subset Q_{ca} = \phi_c(Q_a)$ and $Q_v \subset \phi_c(Q_{a'})$, and these are disjoint because $Q_a \cap Q_{a'} = \varnothing$ by (H1) and ϕ_c is injective. For (c), write $v = uat$; then $Q_v \subset \phi_u(Q_a)$ while $K_u = \phi_u(K_\epsilon)$ with $K_\epsilon \cap Q_a = \varnothing$ by (H4), and injectivity of ϕ_u separates them. (d) combines (b) and (c): incomparable addresses have disjoint cylinders at the first difference, and comparable distinct addresses fall under (c) (every end fiber E_ω lies in $Q_{\omega|m}$ for all m, and two distinct ends are incomparable at some depth). □

Lemma 4.2 (Label-limit lemma)**.** *Let* $\alpha_j \to \alpha$ *in* Ω_∂ *and let* $x_j \in \mathrm{fib}(\alpha_j)$ *with* $x_j \to x$ *in* $\mathbb{C}$. *Then* $x \in \mathrm{fib}(\alpha)$.

Proof. If $\alpha = w\#^\infty$ is a finite address, it is isolated in Ω_∂: any address agreeing with it through position $|w|+1$ (that is, on w followed by $\#$) equals it. Hence $\alpha_j = \alpha$ eventually, and $x \in K_w$ because K_w is closed. If $\alpha = \omega \in \mathtt{A}^{\mathbb{N}}$, fix m. Eventually α_j agrees with ω on the first m symbols; whether α_j is a finite word $v \sqsupseteq \omega|m$ or an end extending $\omega|m$, its fiber lies in the closed cylinder $Q_{\omega|m}$ by Lemma 4.1(a). Hence $x \in Q_{\omega|m}$ for every m, that is, $x \in E_\omega$. □

4.2. **The compactification theorem.**

Theorem 4.3 (Direct-decoration rooted-end compactification)**.** *Under Standing Hypotheses 2.2:*

(i) Ω_∂ *is compact; the finite addresses are isolated, and finite words of increasing length converge to an end exactly by eventual agreement on every fixed prefix.*
(ii) $\mathcal{H}_\partial$ *is a compact subset of* $K(f)$*; under (H3),* $\mathcal{H}_\partial \subset J(f)$. *Every fiber (4.1), (4.2) is a nonempty continuum.*
(iii) *The connected components of* $\mathcal{H}_\partial$ *are exactly the fibers* K_w $(w \in \mathtt{A}^{<\mathbb{N}})$ *and* E_ω $(\omega \in \mathtt{A}^{\mathbb{N}})$.
(iv) *The realization* $\Omega_\partial \ni \alpha \mapsto \mathrm{fib}(\alpha)$ *has closed graph; equivalently, it is upper semicontinuous as a map into the compact subsets of* $\mathcal{H}_\partial$.
(v) *The dynamics acts through actual univalent branches of* f^q*:*

$$g(K_{aw}) = K_w, \qquad g(E_{a\omega}) = E_\omega \qquad (a \in \mathtt{A}), \tag{4.5}$$

each transition having time q *and inverse sheet the deleted first symbol* a.

Proof. (i). The product $\{0,1,\#\}^{\mathbb{N}}$ is compact. The subset (4.3) is defined by the closed conditions "if the k-th symbol is $\#$ then so is the $(k+1)$-st": each violating set is a clopen cylinder, so Ω_∂ is closed, hence compact. Isolation of finite addresses and the stated convergence are immediate in the product topology (see the proof of Lemma 4.2).

(ii). Each Q_w is a closed Jordan disk and $g^{|w|}\colon Q_w \to P$ is a homeomorphism (Lemma 4.1(a)), so each $K_w = \phi_w(K_\epsilon)$ is a continuum, and each E_ω is a nested intersection of continua, hence a nonempty continuum. Every $z \in E_\omega$ satisfies $g^m(z) \in P$ for all $m \geq 0$; the intermediate iterates $f^{mq+r}(z)$, $0 \leq r < q$, lie in the compact union $\bigcup_{r=0}^{q-1} f^r(P)$, so the forward orbit of z is bounded and $z \in K(f)$. The root satisfies $K_\epsilon \subset K(f)$: its points have their f^p-orbits in K_ϵ. For $z \in K_\epsilon$, write an iterate as $mp+i$ with $0 \leq i < p$; then

$$f^{mp+i}(z) \in f^i(K_\epsilon) = C_i.$$

The finite union $\bigcup_{i=0}^{p-1} C_i$ is compact, so the full forward orbit of z is bounded. Complete invariance of $K(f)$ then puts every $K_w = \phi_w(K_\epsilon) \subset f^{-q|w|}(K(f)) = K(f)$. Under (H3), $K(f) = J(f)$.

For compactness of $\mathcal{H}_\partial$, let $x_j \in \mathcal{H}_\partial$ converge to x, and let $\alpha_j \in \Omega_\partial$ be addresses with $x_j \in \mathrm{fib}(\alpha_j)$. By compactness of Ω_∂ pass to a subsequence with $\alpha_j \to \alpha$; Lemma 4.2 gives $x \in \mathrm{fib}(\alpha) \subset \mathcal{H}_\partial$.

(iv). The same argument applied without passing to a subsequence of the x_j proves that the graph $\{(\alpha, x) : x \in \mathrm{fib}(\alpha)\}$ is closed in $\Omega_\partial \times \mathcal{H}_\partial$. Since $\mathcal{H}_\partial$ is compact, a closed graph is equivalent to upper semicontinuity of the set-valued realization.

(iii). Each fiber is connected by (ii), and distinct fibers are disjoint by Lemma 4.1(d). It remains to show that no connected subset of $\mathcal{H}_\partial$ meets two fibers. For a finite word u, consider

$$\mathcal{H}_\partial[u] := \bigcup_{v \sqsupseteq u} K_v \cup \bigcup_{\omega \sqsupseteq u} E_\omega.$$

We claim $\mathcal{H}_\partial[u]$ is clopen in $\mathcal{H}_\partial$. First, $\mathcal{H}_\partial[u] = \mathcal{H}_\partial \cap Q_u$: fibers with address extending u lie in Q_u by Lemma 4.1(a), fibers with address incomparable to u miss Q_u by (b), and fibers K_v with v a strict prefix of u miss Q_u by (c). Hence $\mathcal{H}_\partial[u]$ is closed. Second, its complement $\mathcal{H}_\partial \setminus \mathcal{H}_\partial[u]$ is the union of the fibers over the addresses not extending u; these addresses form a clopen, hence compact, subset $N_u \subset \Omega_\partial$, and the union of fibers over any compact address set is closed: the graph of the realization restricted to N_u is a closed subset of the compact space $N_u \times \mathcal{H}_\partial$ ((iv)), hence compact, and its projection to $\mathcal{H}_\partial$ is therefore compact. Thus $\mathcal{H}_\partial[u]$ is clopen.

Now let $C \subset \mathcal{H}_\partial$ be connected and meet the fibers of two distinct addresses $\alpha \neq \beta$. If α, β first differ at position k (both defined and unequal there), the clopen sets $\mathcal{H}_\partial[\alpha|_k]$ and its complement separate the two fibers, so C cannot meet both. Otherwise one address is a finite word u and the other strictly extends it; then β extends ua for some $a \in \mathtt{A}$, the fiber of β lies in the clopen set $\mathcal{H}_\partial[ua]$, and K_u lies in its complement by Lemma 4.1(c). In both cases C is disconnected by a clopen partition, a contradiction. So the components of $\mathcal{H}_\partial$ are exactly the fibers.

(v). From $g \circ \phi_a = \mathrm{id}_P$ we get $g \circ \phi_{aw} = \phi_w$, hence $g(K_{aw}) = K_w$. Since $g\colon Q_a \to P$ is a homeomorphism (H2), it commutes with the nested intersection (4.2):

$$g(E_{a\omega}) = g\Big(\bigcap_m Q_{(a\omega)|m}\Big) = \bigcap_m g\big(Q_{a(\omega|m-1)}\big) = \bigcap_m Q_{\omega|m-1} = E_\omega.$$

Each transition is the restriction of f^q to a subset of Q_a, and the deleted first symbol a records the inverse sheet. □

4.3. **The component quotient and partial shift.**

Theorem 4.4 (Component quotient)**.** *Define $\pi_\partial\colon \mathcal{H}_\partial \to \Omega_\partial$ by $\pi_\partial(x) = \alpha \iff x \in \mathrm{fib}(\alpha)$. Then π_∂ is a well-defined continuous closed surjection whose fibers are precisely the connected components of $\mathcal{H}_\partial$. Consequently the component space of $\mathcal{H}_\partial$, with the quotient topology, is homeomorphic to Ω_∂:*

$$\mathcal{H}_\partial / \sim_{component} \cong \Omega_\partial.$$

In particular the component decomposition of $\mathcal{H}_\partial$ is upper semicontinuous.

Proof. Well-definedness and surjectivity hold because the nonempty fibers are pairwise disjoint and cover $\mathcal{H}_\partial$ (Lemma 4.1(d), Theorem 4.3(ii)). For continuity, let $x_j \to x$ in $\mathcal{H}_\partial$ and write $\alpha_j = \pi_\partial(x_j)$, $\alpha = \pi_\partial(x)$. If $\alpha_j \not\to \alpha$, choose a neighborhood V of α that infinitely many α_j avoid; compactness of $\Omega_\partial \setminus V$ gives a subsequence converging to some $\beta \neq \alpha$. Lemma 4.2 then

puts $x \in \mathrm{fib}(\beta)$, while $x \in \mathrm{fib}(\alpha)$, contradicting disjointness of fibers. Hence $\alpha_j \to \alpha$ and π_∂ is continuous. A continuous map from a compact space to a Hausdorff space is closed; a closed continuous surjection is a quotient map, and its fibers are the components by Theorem 4.3(iii). Upper semicontinuity of the decomposition is exactly closedness of the quotient map with compact fibers. □

Theorem 4.5 (Partial-shift realization)**.** *For $a \in \mathtt{A}$ put $H_a = \mathcal{H}_\partial \cap Q_a$. Then:*

(i) *$H_a = \phi_a(\mathcal{H}_\partial)$, and H_0, H_1 are disjoint clopen subsets of $\mathcal{H}_\partial$ with*

$$\mathcal{H}_\partial \setminus K_\epsilon = H_0 \sqcup H_1;$$

(ii) *$g|_{H_a} \colon H_a \to \mathcal{H}_\partial$ is a homeomorphism with inverse ϕ_a; hence*

$$g \colon \mathcal{H}_\partial \setminus K_\epsilon \longrightarrow \mathcal{H}_\partial$$

is a (trivial) two-sheeted covering with clopen sheets H_0, H_1;

(iii) *the component quotient semiconjugates this covering to the first-symbol deletion shift:*

$$\pi_\partial \circ g = \sigma_{\mathrm{pre}} \circ \pi_\partial \qquad \text{on } \mathcal{H}_\partial \setminus K_\epsilon.$$

Proof. (i) The identities $\phi_a(K_w) = K_{aw}$ and $\phi_a(E_\omega) = E_{a\omega}$ show $\phi_a(\mathcal{H}_\partial) = \bigcup_w K_{aw} \cup \bigcup_\omega E_{a\omega} = \mathcal{H}_\partial[a]$, which equals $\mathcal{H}_\partial \cap Q_a$ and is clopen by the proof of Theorem 4.3(iii). Every fiber other than the root fiber has an address beginning with 0 or 1, so $\mathcal{H}_\partial \setminus K_\epsilon = H_0 \sqcup H_1$, and $H_0 \cap H_1 = \varnothing$ by (H1).

(ii) On $H_a \subset Q_a$ the map g is the restriction of the homeomorphism $g \colon Q_a \to P$ of (H2), with inverse ϕ_a; by (i), ϕ_a carries $\mathcal{H}_\partial$ onto H_a, so $g|_{H_a}$ is a homeomorphism onto $\mathcal{H}_\partial$. The two sheets are clopen and disjoint, so the restriction of g to their union is a trivial two-sheeted covering of $\mathcal{H}_\partial$.

(iii) On H_a, the fiber of address $a\beta$ maps onto the fiber of address β by (4.5); this is exactly the semiconjugacy identity. □

4.4. **A visual atlas of the binary geometry.** The next two plates summarize the address geometry proved above and the two different parent operations formalized below. They are schematic guides: a drawn blob represents a connected fiber, while an address arrow records only the operation stated in its label.

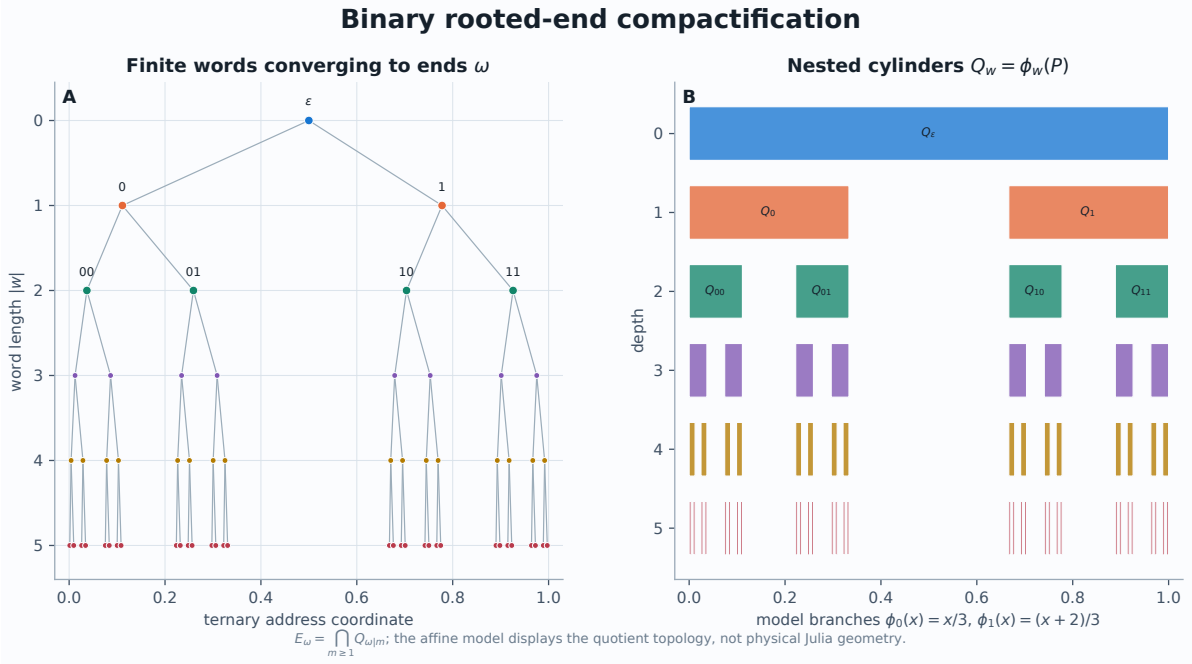


FIGURE 1. Binary cylinders, finite fibers, and end fibers. For an infinite address ω, the nested intersection $E_\omega = \bigcap_{m\geq 1} Q_{\omega|m}$ is a nonempty continuum, not a point. Cantor space appears only at the component-quotient level: $\mathcal{H}_\infty/\{\text{components}\} \cong \mathtt{A}^{\mathbb{N}} = \{0,1\}^{\mathbb{N}}$. The full component quotient is instead $\mathcal{H}_\partial/\{\text{components}\} \cong \Omega_\partial$, which also contains the isolated finite addresses.

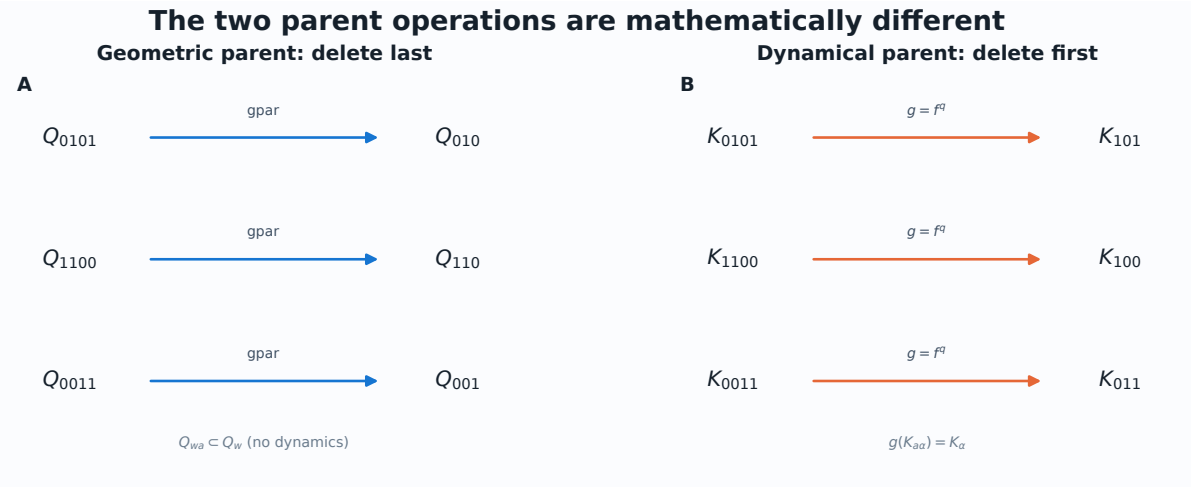


FIGURE 2. The two parent operations must not be conflated. The geometric parent $\mathrm{gpar}(wa) = w$ deletes the last symbol and records only the nesting $Q_{wa} \subset Q_w$; it is not induced by a dynamical map. The dynamical parent $\sigma_{\mathrm{pre}}(a\alpha) = \alpha$ deletes the first symbol and is realized on non-root fibers by $g = f^q$. The root fiber has the separate transition $\tau_{\mathrm{root}} \colon K_\epsilon \mapsto C_{q \bmod p}$ rather than an extension of σ_{pre}.

4.5. **The three transition maps.** The two arrows are easiest to compare when their geometries are kept in separate panels.

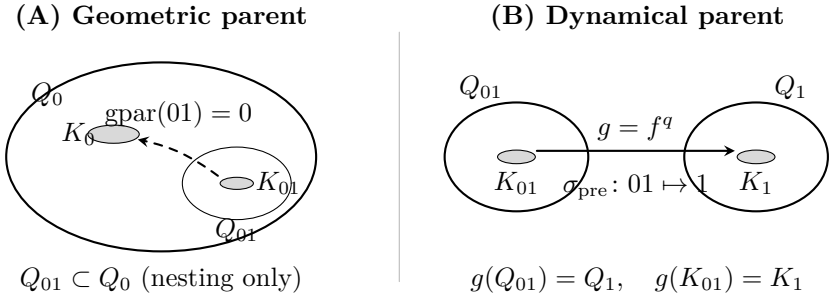


FIGURE 3. The two parent maps at address 01. (A) Last-symbol deletion records the inclusion $Q_{01} = \phi_0(Q_1) \subset Q_0$ and has no dynamical content. (B) First-symbol deletion is the actual f^q-transition. Address fibers use word subscripts K_w; terminal-cycle components use the distinct notation C_i.

The geometric and dynamical parent maps are

$$\mathrm{gpar}(wa) = w, \qquad \sigma_{\mathrm{pre}}(a\alpha) = \alpha. \tag{4.6}$$

The *geometric parent* gpar deletes the *last* symbol of a nonempty finite word. It is realized by cylinder inclusion, $Q_{wa} \subset Q_w$, and by *no* dynamical map: there is in general no iterate of f carrying K_{wa} onto K_w. It has no action on ends, which have no last symbol.

The *dynamical parent* σ_{pre} deletes the *first* symbol. It is defined on all non-root finite addresses and on *all* ends, and it is realized by the two-branch covering of Theorem 4.5, with transition time q and the deleted symbol as inverse sheet. Concretely, at address 01: $\mathrm{gpar}(01) = 0$ because $Q_{01} = \phi_0(Q_1) \subset Q_0$, while $\sigma_{\mathrm{pre}}(01) = 1$ because $g(K_{01}) = K_1$. The two maps are not equal. On a word of length at least two they agree precisely for constant words; for example, they agree on 00 but differ on 01.

The dynamical parent σ_{pre} is undefined at the empty word. The root has its own transition, coming from the terminal cycle of Proposition 2.1:

$$\tau_{\mathrm{root}}\colon \quad g(K_\epsilon) = f^q(K_\epsilon) = C_{q \bmod p}. \tag{4.7}$$

This arrow lands on a component of the terminal cycle; its target is indexed by the cycle rather than by an address, and no positional claim about $C_{q \bmod p}$ relative to $\mathcal{H}_\partial$ is made beyond Proposition 2.1. It is *not* an extension of σ_{pre}, and no assertion is made that the image escapes $\mathcal{H}_\partial$ permanently (Standing scope (N1)).

4.6. **Naturality under root enlargement.** For a compact connected R with $K_\epsilon \subset R \subset P \setminus (Q_0 \cup Q_1)$, write

$$\mathcal{H}_\partial(R) = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} \phi_w(R) \;\cup\; \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega$$

for the compactum with root R; thus $\mathcal{H}_\partial = \mathcal{H}_\partial(K_\epsilon)$.

Theorem 4.6 (Root-gap enlargement naturality)**.** *Let*

$$K_\epsilon \subset S \subset S^+ \subset P \setminus (Q_0 \cup Q_1) \tag{4.8}$$

be compact connected roots. Then for each $R \in \{S, S^+\}$: the separation, compactness, component, quotient, and partial-shift conclusions of Lemmas 4.1 and 4.2 and Theorems 4.3 to 4.5 hold for $\mathcal{H}_\partial(R)$ with root fiber R. Moreover,

$$\mathcal{H}_\partial(R) \subset K(f) \iff R \subset K(f),$$

and under (H3) the same equivalence holds with $J(f)$. The inclusions $\phi_w(S) \subset \phi_w(S^+)$ commute with g, and the end fibers are unchanged.

Proof. The cited proofs use the root only through compactness, connectedness, containment in P, and disjointness from $Q_0 \cup Q_1$, all of which hold by (4.8). If $R \subset K(f)$, complete invariance gives $\phi_w(R) \subset f^{-q|w|}(K(f)) = K(f)$, and the end fibers lie in $K(f)$ by the bounded-orbit argument of Theorem 4.3. The converse follows because R is the root fiber. Finally, injectivity of ϕ_w gives the stated inclusions, $g \circ \phi_a = \mathrm{id}$ gives commutativity, and the end fibers depend only on P, Q_0, Q_1. □

Fullness and cell conclusions require the enlarged root to be full and compactly contained in G_0; see Corollary 5.7.

5. Fullness, cells, and replacement

5.1. Full fibers and connected complement.

Theorem 5.1 (Full fibers and nonseparation)**.** *Every fiber K_w and E_ω of Theorem 4.3 is a full continuum, and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected. Moreover, for every Jordan domain U with $P \Subset U$, the relative complement $U \setminus \mathcal{H}_\partial$ is connected.*

Proof. The root K_ϵ is full [DH85]. Each $\phi_w \colon P \to Q_w$ is an orientation-preserving homeomorphism of closed Jordan disks, so $K_w = \phi_w(K_\epsilon)$ is full by Lemma 3.3(a). For an end,

$$\widehat{\mathbb{C}} \setminus E_\omega = \bigcup_{m \geq 1} \left(\widehat{\mathbb{C}} \setminus Q_{\omega|m}\right)$$

is an increasing union of connected sets containing ∞, hence connected; so E_ω is full.

For $m \geq 0$ define the approximants

$$\mathcal{H}_\partial^{(m)} = \bigcup_{|u|<m} K_u \;\cup \bigcup_{|w|=m} Q_w. \tag{5.1}$$

Each term is full, and the terms of $\mathcal{H}_\partial^{(m)}$ are pairwise disjoint: equal-depth cylinders are disjoint (Lemma 4.1(b)), distinct finite fibers are disjoint ((d)), and for $|u| < m = |w|$ the word u is either incomparable to w (disjoint cylinders) or a strict prefix of w (then $K_u \cap Q_w = \varnothing$ by (c)). The sets (5.1) decrease in m: each K_u with $|u| = m$ lies in Q_u, and each Q_w with $|w| = m+1$ lies in $Q_{w|m}$.

Their intersection is exactly $\mathcal{H}_\partial$. Indeed $\mathcal{H}_\partial \subset \mathcal{H}_\partial^{(m)}$ for every m (a fiber with address of length $\geq m$ or an end fiber lies in its depth-m cylinder). Conversely, let $x \in \bigcap_m \mathcal{H}_\partial^{(m)}$. If x lies in some finite fiber K_u we are done. Otherwise, for every m there is a unique word w_m of length m with $x \in Q_{w_m}$ (uniqueness by disjointness of equal-depth cylinders), and $x \in Q_{w_{m+1}} \subset Q_{w_{m+1}|m}$ forces $w_{m+1}|m = w_m$. The words w_m therefore converge to an end ω with $x \in \bigcap_m Q_{\omega|m} = E_\omega$.

Lemma 3.2 applied to the decreasing approximants (5.1) proves that $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected; its localized clause, applied with the exterior full continuum $\widehat{\mathbb{C}} \setminus U$ adjoined to each approximant, proves that $U \setminus \mathcal{H}_\partial$ is connected. □

5.2. **The end compactum.**

Corollary 5.2 (Cantor-indexed end compactum)**.** *Let*

$$\mathcal{H}_\infty = \bigcup_{\omega\in\mathtt{A}^{\mathbb{N}}} E_\omega.$$

Then

$$\mathcal{H}_\infty = \bigcap_{m\ge 1} \bigcup_{|w|=m} Q_w, \tag{5.2}$$

$\mathcal{H}_\infty$ is compact, its connected components are precisely the end fibers E_ω, and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\infty$ is connected. The component quotient restricts to a continuous closed surjection $\mathcal{H}_\infty \to \mathtt{A}^{\mathbb{N}} = \{0,1\}^{\mathbb{N}}$, so the component space of $\mathcal{H}_\infty$ is the Cantor space, and

$$g\colon \mathcal{H}_\infty \longrightarrow \mathcal{H}_\infty$$

is a trivial two-sheeted covering of $\mathcal{H}_\infty$ over itself, semiconjugate under the quotient to the one-sided binary shift, realized by the two univalent branches ϕ_0, ϕ_1 of f^{-q}.

Proof. For (5.2): every E_ω lies in $Q_{\omega|m}$ for all m, giving one inclusion. Conversely, a point x of the right-hand side lies, for every m, in a unique depth-m cylinder Q_{w_m}, and uniqueness forces $w_{m+1}|m = w_m$ as in Theorem 5.1; hence $x \in E_\omega$ for $\omega = \lim w_m$. No finite fiber meets the right-hand side: for $m > |u|$, K_u misses every depth-m cylinder (Lemma 4.1(c) and (b)). Thus (5.2) holds, and $\mathcal{H}_\infty$ is an intersection of compacta, hence compact.

The levels $\bigcup_{|w|=m} Q_w$ are finite unions of pairwise disjoint full continua and decrease in m, so Lemma 3.2 makes $\widehat{\mathbb{C}} \setminus \mathcal{H}_\infty$ connected. Components: the sets $\mathcal{H}_\infty \cap Q_u$ are clopen in $\mathcal{H}_\infty$ (they are the restrictions of the clopen sets $\mathcal{H}_\partial[u]$ of Theorem 4.3), and the argument of Theorem 4.3(iii) applies verbatim with only end addresses present. The quotient statement restricts Theorem 4.4; $\mathtt{A}^{\mathbb{N}} \subset \Omega_\partial$ is closed, and the restricted map is again a continuous closed surjection. Finally $\phi_a(\mathcal{H}_\infty) = \mathcal{H}_\infty \cap Q_a$ and $g(E_{a\omega}) = E_\omega$ give the two-sheeted covering of $\mathcal{H}_\infty$ over itself and the semiconjugacy to the full one-sided shift, exactly as in Theorem 4.5; here no root is removed, since σ_{pre} is total on ends. □

Remark 5.3. $\mathcal{H}_\infty$ is the statement most directly comparable to a "Cantor family of continua": a compact nonseparating union of full continua whose component space is $\{0,1\}^{\mathbb{N}}$ and on which the binary shift is realized by two actual univalent inverse branches. It does not identify the E_ω as small Julia sets (Standing scope (N2)); it realizes exactly the topological and dynamical skeleton.

5.3. **The gap-cell atlas.** Recall G_0, the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing K_ϵ (Standing Hypotheses 2.2(H4)). By Lemma 3.4 choose once and for all a closed Jordan disk D with

$$K_\epsilon \subset \operatorname{int} D \subset D \Subset G_0, \tag{5.3}$$

and put $D_w = \phi_w(D)$.

Theorem 5.4 (Gap-cell atlas)**.** *The closed Jordan cells D_w are pairwise disjoint—including for ancestor–descendant pairs—and satisfy*

$$K_w \subset \operatorname{int} D_w, \qquad D_w \Subset \operatorname{int} Q_w \setminus (Q_{w0} \cup Q_{w1}), \tag{5.4}$$

$$D_w \cap \mathcal{H}_\partial = K_w, \qquad g(D_{aw}) = D_w. \tag{5.5}$$

If $S_w \Subset \operatorname{int} D_w$ is compact and connected, then

$$\operatorname{Fill}(S_w) \Subset \operatorname{int} D_w. \tag{5.6}$$

Proof. By (5.3) and (H2), ϕ_w carries $\operatorname{int} P$ to $\operatorname{int} Q_w$, Q_a to Q_{wa}, and D to D_w, which proves (5.4). For disjointness: incomparable words have disjoint cylinders containing their cells; and if $v = uat$ is a strict descendant of u, then D_v lies in the child cylinder Q_{ua} while D_u misses both children of Q_u by (5.4).

First identity of (5.5): fibers with address strictly extending w lie in the children $Q_{w0} \cup Q_{w1}$, which D_w misses; fibers with address incomparable to w miss $Q_w \supset D_w$; fibers K_u with u a strict prefix of w miss $Q_{w|_{|u|+1}} \supset D_w$ by Lemma 4.1(c); end fibers extending w lie in a child of Q_w. What remains is $K_w \subset \operatorname{int} D_w$. The second identity follows from $g \circ \phi_a = \mathrm{id}$.

For (5.6): a collar of ∂D_w together with the exterior of D_w is a connected subset of the unbounded component of $\mathbb{C} \setminus S_w$; every bounded complementary component of S_w is therefore contained away from that collar, inside $\operatorname{int} D_w$. □

5.4. **Cellwise replacement at arbitrary addresses.** Replacement may be performed at an arbitrary set of finite addresses.

Theorem 5.5 (Cellwise replacement at arbitrary addresses)**.** *Let $F \subset \mathtt{A}^{<\mathbb{N}}$ be an arbitrary—possibly infinite—set of finite addresses, and for each $w \in F$ let L_w be a* nonempty *compact connected set with*

$$K_w \subset L_w \Subset \operatorname{int} D_w. \tag{5.7}$$

Put $\widetilde{K}_w = L_w$ for $w \in F$ and $\widetilde{K}_w = K_w$ otherwise, and

$$\widetilde{\mathcal{H}} = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} \widetilde{K}_w \;\cup\; \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega. \tag{5.8}$$

Then:

(i) *$\widetilde{\mathcal{H}}$ is compact; its connected components are exactly the sets $\widetilde{K}_w$ ($w \in \mathtt{A}^{<\mathbb{N}}$) and E_ω ($\omega \in \mathtt{A}^{\mathbb{N}}$); the realization $\alpha \mapsto \widetilde{\operatorname{fib}}(\alpha)$ is upper semicontinuous; and the component projection $\widetilde{\pi}\colon \widetilde{\mathcal{H}} \to \Omega_\partial$ is a continuous closed surjection.*

(ii) *If in addition every L_w ($w \in F$) is full, then $\widehat{\mathbb{C}} \setminus \widetilde{\mathcal{H}}$ is connected, and for every Jordan domain U with $P \Subset U$ the relative complement $U \setminus \widetilde{\mathcal{H}}$ is connected.*

Proof. Define the modified approximants

$$\widetilde{\mathcal{H}}^{(m)} = \bigcup_{|u|<m} \widetilde{K}_u \;\cup\; \bigcup_{|w|=m} Q_w. \tag{5.9}$$

At each fixed m this is a *finite* union—fewer than 2^m finite-address terms and exactly 2^m cylinders—regardless of the cardinality of F. The terms are pairwise disjoint: distinct finite-address terms lie in the pairwise disjoint cells D_u (both K_u and L_u lie in $\operatorname{int} D_u$ by (5.4) and (5.7)); a term $\widetilde{K}_u$ with $|u| < m$ and a cylinder Q_w with $|w| = m$ are disjoint because either u is a strict prefix of w, in which case Q_w lies in a child of Q_u, which $D_u \supset \widetilde{K}_u$ misses by (5.4), or u and w are incomparable, in which case already $Q_u \cap Q_w = \varnothing$. The sets (5.9) decrease in m: each $\widetilde{K}_u$ with $|u| = m$ lies in $\operatorname{int} D_u \subset Q_u$, and each depth-$(m{+}1)$ cylinder lies in its parent.

The intersection of the modified approximants is exactly $\widetilde{\mathcal{H}}$: one inclusion is clear as before, and if $x \in \bigcap_m \widetilde{\mathcal{H}}^{(m)}$ lies in no finite-address term $\widetilde{K}_u$, then at every depth it lies in a unique cylinder, the words nest, and x lies in the corresponding end fiber. Compactness of $\widetilde{\mathcal{H}}$ follows, each approximant being compact.

For the component structure, upper semicontinuity, and the quotient in (i), the arguments of Theorems 4.3 and 4.4 apply verbatim once their two inputs are re-verified for the modified fibers. First, the separations: distinct finite-address terms lie in disjoint cells; such a term $\widetilde{K}_u$

misses every cylinder Q_v with v strictly extending u (it lies in D_u, which misses the children of Q_u) and every cylinder incomparable to u; hence distinct modified fibers are pairwise disjoint, and the clopen sets $\widetilde{\mathcal{H}} \cap Q_u$ again consist exactly of the modified fibers with address extending u. Second, the label-limit lemma: if $\alpha_j \to \alpha$ and $x_j \in \widetilde{\mathrm{fib}}(\alpha_j) \to x$, then for finite α the labels are eventually constant and $\widetilde{K}_\alpha$ is closed, while for an end α every modified fiber with address extending $\alpha|m$ lies in $Q_{\alpha|m}$ (finite-address terms via $\widetilde{K}_v \subset D_v \subset Q_v \subset Q_{\alpha|m}$), so $x \in E_\alpha$. With these two inputs, the clopen-partition components argument, the closed-graph upper semicontinuity, and the continuity-closedness of the component projection repeat word for word.

For (ii), every term of (5.9) is full (L_w by hypothesis, K_u and Q_w as before), so Lemma 3.2 and its localized clause apply to the decreasing compacta, with the exterior continuum $\widehat{\mathbb{C}} \setminus U$ adjoined for the relative statement. □

Remark 5.6. No uniform diameter estimate, no coherence between the L_w at different addresses, and no dynamical compatibility is required in Theorem 5.5; correspondingly, no equivariance is asserted for the replaced compactum—$g(L_{aw})$ need not equal L_w. Equivariance returns for coherent families $L_w = \phi_w(S)$, which is the content of Corollary 5.7. For nonfull replacements, part (i) survives while part (ii) can fail: a Jordan curve joined to K_w inside D_w creates a bounded complementary component. The containment $K_w \subset L_w$ is not used in either conclusion and may be dropped, provided nonemptiness is retained so that the component projection remains surjective.

5.5. **Full root-gap naturality.**

Corollary 5.7 (Full root-gap naturality)**.** *Let S be a nondegenerate full continuum with*

$$K_\epsilon \subset S \Subset G_0,$$

and put $S_w = \phi_w(S)$. Then:

- (i) *$\mathcal{H}_\partial(S) = \bigcup_w S_w \cup \bigcup_\omega E_\omega$ is compact, its components are exactly the S_w and E_ω, the component projection onto Ω_∂ is a continuous closed surjection, and $g\colon \mathcal{H}_\partial(S) \setminus S \to \mathcal{H}_\partial(S)$ is a trivial two-sheeted covering with $g(S_{aw}) = S_w$;*
- (ii) *every S_w is full, $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial(S)$ is connected, and $U \setminus \mathcal{H}_\partial(S)$ is connected for every Jordan domain U with $P \Subset U$;*
- (iii) *there is a gap-cell atlas for S: a closed Jordan disk D_S with $S \subset \operatorname{int} D_S \subset D_S \Subset G_0$ whose transported cells $D_{S,w} = \phi_w(D_S)$ satisfy all conclusions of Theorem 5.4 with S_w in place of K_w;*
- (iv) *$\mathcal{H}_\partial(S) \subset K(f)$ if and only if $S \subset K(f)$; under (H3) the same holds with $J(f)$.*

Proof. Apply Theorem 4.6 with root S for (i) and (iv), and use $g \circ \phi_{aw} = \phi_w$ for equivariance. The proofs of Theorems 5.1 and 5.4 apply with the full root $S \Subset G_0$, giving (ii) and (iii). □

6. Protected carriers

Finitely many protected accesses at the root can be transported to selected fibers and joined by a full carrier. Finiteness enters through the connector tree, not through the replacement theorem.

6.1. **Connector tools.**

Lemma 6.1 (One-point tree attachments are full)**.** *If a finite embedded plane tree meets each of finitely many pairwise disjoint full continua in exactly one point and otherwise misses them, the union is a full continuum.*

Proof. A finite embedded plane tree is nonseparating. Attach the full continua one at a time. At each step the two nonseparating continua meet in a connected singleton, and the Janiszewski separation lemma [Kur68, §61] implies that their union is nonseparating. Induction proves that the final compact connected set has connected complement. □

Lemma 6.2 (One-ended slit lemma)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $\delta_1, \dots, \delta_s \colon [0,1] \to \overline{\Omega}$ be pairwise disjoint Jordan arcs with $\delta_i(0) \in \partial\Omega$ and $\delta_i((0,1]) \subset \Omega$. Then*

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i((0,1])$$

is path connected. Moreover,

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i((0,1))$$

is path connected and contains every free tip $\delta_i(1)$.

Proof. For one slit, begin with a polygonal path in Ω. It misses an initial segment at the boundary endpoint and, after a perturbation, meets the remaining compact subarc finitely many times. A thin rectangular neighborhood of this tame planar arc [Moi77, Chap. 10, Theorem 10.8] allows each crossing to be detoured along its two sides and around the free endpoint. Induction over the disjoint slits proves the first assertion.

Let $O = \Omega \setminus \bigcup_i \delta_i((0,1])$. A sufficiently small disk about each free tip meets no other slit and, in a tame-arc chart, contains an arc from the tip into O whose interior misses its own slit. Attaching these finitely many arcs to the path-connected set O proves the second assertion. □

6.2. **Crosscut chains.** Finally we fix the standard crosscut and prime-end conventions; see [Pom92, §2.4]. A *crosscut* of the complement of a full continuum A is a Jordan arc whose interior lies in $\widehat{\mathbb{C}} \setminus A$ and whose two distinct endpoints lie on ∂A. A *crosscut chain* is a sequence

$$(\gamma_n, D_n)_{n \geq 1}$$

of pairwise disjoint crosscuts γ_n with selected sides D_n, where:

(C1) D_n is a selected component of $\widehat{\mathbb{C}} \setminus (A \cup \gamma_n)$;

(C2) the chain is *nulled*:

$$\overline{D_{n+1}} \setminus A \subset D_n \qquad (n \geq 1); \tag{6.1}$$

(C3) $\operatorname{diam} \gamma_n \to 0$.

Condition (6.1) contains the usual null-chain relation: since the open arc γ_{n+1}° lies in $\overline{D_{n+1}} \setminus A$, the later crosscuts satisfy $\gamma_{n+1}^{\circ} \subset D_n$, so for $n \geq 2$ every crosscut interior lies in D_1. The chain defines a prime end of $\widehat{\mathbb{C}} \setminus A$ with impression $\bigcap_n \overline{D_n}$. We call $\overline{D_n} \setminus A$ the *relative collar tail* at level n. Relative collar tails of different chains may have closures meeting inside A—in particular, overlapping prime-end impressions are allowed, and no local connectivity of A is ever assumed.

Lemma 6.3 (Protected crosscut survival and order)**.** *Let A be a full continuum, let $H \supset A$ be a full continuum, and let (γ_n, D_n) be a crosscut chain of $\widehat{\mathbb{C}} \setminus A$. If*

$$(H \setminus A) \cap (\overline{D_1} \setminus A) = \varnothing, \tag{6.2}$$

then the crosscuts $(\gamma_n)_{n \geq 2}$ with their selected sides form a prime-end chain in $\widehat{\mathbb{C}} \setminus H$; the endpoints remain boundary points and the selected sides are the same domains D_n. A finite family of such chains with pairwise disjoint relative terminal collar tails induces distinct prime

ends of $\widehat{\mathbb{C}} \setminus H$ with unchanged cyclic order. Here the preserved cyclic order is that of the transported crosscut family in the one ambient complement $\widehat{\mathbb{C}} \setminus H$; no identification of prime-end circles across different atoms is asserted.

Proof. For $n \geq 2$, (6.1) gives $\gamma_n^\circ \subset D_{n-1} \subset D_1$, so $\gamma_n^\circ \subset \overline{D_1} \setminus A$ misses H by (6.2), while the two distinct endpoints of γ_n lie in $A \subset H$. Every neighborhood of either endpoint meets the crosscut interior, so both endpoints remain on the boundary of the enlarged complement; thus γ_n remains a crosscut of $\widehat{\mathbb{C}} \setminus H$.

Moreover, $D_n \subset \overline{D_n} \setminus A \subset D_1$ for $n \geq 2$, so $D_n \cap H = \varnothing$. As a component of $\widehat{\mathbb{C}} \setminus (A \cup \gamma_n)$, its boundary satisfies $\partial D_n \subset A \cup \gamma_n \subset H \cup \gamma_n$. Hence D_n itself is a component of $\widehat{\mathbb{C}} \setminus (H \cup \gamma_n)$: it is connected and both open and closed relative to that complement. Therefore the selected sides, their nesting relation, and the crosscut diameters are unchanged.

Equivalent chains eventually divide one another; pairwise disjoint relative terminal collar tails prevent that, so the induced prime ends are distinct. For a finite family, choose one deep oriented crosscut from each chain: prime-end cyclic order is the separation order of these crosscuts, and neither the crosscuts nor their selected sides changed, so every ordered triple retains its orientation. □

6.3. **Connector data.** Throughout, U is a Jordan domain, $A_1, \dots, A_s \Subset U$ are pairwise disjoint full continua (the *atoms*), and $Z \Subset U \setminus \bigcup_i A_i$ is a compact *forbidden set*. For every protected access to A_i a crosscut chain $\Gamma_{i,a} = (\gamma_{i,a,n}, D_{i,a,n})_{n\geq 1}$ of $\widehat{\mathbb{C}} \setminus A_i$ is chosen, truncated so that the relative terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$ are pairwise disjoint, lie in U, and miss Z and the other atoms. The closures of two tails may meet inside their common atom: overlapping prime-end impressions are allowed.

Definition 6.4 (Connector datum)**.** A *connector datum* consists of the preceding data, pairwise disjoint Jordan port arcs α_i satisfying

$$\alpha_i \cap A_i = \{x_i\} = \{\alpha_i(0)\}, \qquad \alpha_i((0,1]) \cap \Big(Z \cup \bigcup_j A_j \cup \bigcup_{j,a} \big(\overline{D_{j,a,1}} \setminus A_j\big) \Big) = \varnothing,$$

and an embedded finite tree $G \subset U$ joining the outer port endpoints $\alpha_i(1)$ while missing the same protected and forbidden sets, subject to the incidence conditions

$$G \cap \alpha_i = \{\alpha_i(1)\} \quad (1 \leq i \leq s), \qquad T_{\mathfrak{R}} := G \cup \bigcup_i \alpha_i \text{ is a finite embedded plane tree.} \tag{6.3}$$

The planar cyclic order of half-edges at every vertex of $T_{\mathfrak{R}}$ is retained. The port points x_i are required not to be impressions of the protected chains.

Given a connector datum $\mathfrak{R}$, define the *carrier*

$$H_{\mathfrak{R}} = T_{\mathfrak{R}} \cup \bigcup_i A_i = G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i. \tag{6.4}$$

The *attachment trace* of a carrier $H \supset A_i$ at the atom A_i is the compact set $\overline{H \setminus A_i} \cap A_i$.

Theorem 6.5 (Connector realization)**.** *Every connector datum has a carrier $H_{\mathfrak{R}}$ with the following properties:*

(i) *$H_{\mathfrak{R}} \Subset U$ is compact, connected, and full;*
(ii) *it contains every A_i, misses Z, and has singleton attachment trace: $\overline{H_{\mathfrak{R}} \setminus A_i} \cap A_i = \{x_i\}$ for every i;*
(iii) *every protected crosscut chain remains a prime-end chain in $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$;*

(iv) *the survival map on the protected prime ends is injective and preserves the cyclic order of the transported family in* $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$*;*
(v) *no local connectivity of any* A_i *is required.*

Proof. By (6.3), $T_{\mathfrak{R}}$ is a finite embedded plane tree, and it meets each atom A_i exactly in the singleton $\{x_i\}$: the port α_i meets A_i exactly there, the other ports miss A_i by the datum, and G misses every atom. Lemma 6.1 applied to $T_{\mathfrak{R}}$ and the atoms gives (i) for the union (6.4), and the datum gives the containment and avoidance parts of (ii). For the attachment trace, $H_{\mathfrak{R}} \setminus A_i$ is the union of $T_{\mathfrak{R}} \setminus \{x_i\}$ with the other atoms; its closure adds only x_i, and the other atoms are closed and disjoint from A_i, so $\overline{H_{\mathfrak{R}} \setminus A_i} \cap A_i = \{x_i\}$.

For a protected chain at A_i, the set $H_{\mathfrak{R}} \setminus A_i$ misses the relative tail $\overline{D_{i,a,1}} \setminus A_i$ by the datum, verifying (6.2); Lemma 6.3 then proves (iii) and (iv), including the stated reading of cyclic order. The argument uses crosscut chains only, so no Euclidean boundary extension of an atom is assumed, giving (v). □

Let Q be the union of all atoms, all relative terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$, and the forbidden set Z, and put

$$\Omega_{\text{free}} = U \setminus Q. \tag{6.5}$$

This Q is compact. Although a relative tail need not be closed by itself,

$$A_i \cup \left(\overline{D_{i,a,1}} \setminus A_i\right) = A_i \cup \overline{D_{i,a,1}}$$

is compact, and the inventory is finite. Consequently Ω_{free} is an open plane domain. A component Ω_0 of Ω_{free} has *simultaneous ports* if there are pairwise disjoint Jordan arcs $\alpha_i \colon [0,1] \longrightarrow \overline{\Omega_0}$ with $\alpha_i \cap A_i = \{\alpha_i(0)\} = \{x_i\}$, $x_i \in \partial\Omega_0$ not an impression of a protected chain, and $\alpha_i((0,1]) \subset \Omega_0$.

Theorem 6.6 (Free-component criterion)**.** *A connector datum exists if and only if one component of* Ω_{free} *has simultaneous ports to every required atom.*

Proof. Necessity: the connector tree $T_{\mathfrak{R}}$ with its atoms deleted is connected and lies in Ω_{free}, so all its port interiors lie in one component.

Sufficiency: choose the port arcs $\alpha_i \colon [0,1] \to \overline{\Omega_0}$, with $\alpha_i(0) = x_i \in \partial\Omega_0$ and $\alpha_i((0,1]) \subset \Omega_0$. Their interiors $\alpha_i((0,1])$ are finitely many disjoint one-ended slits of the domain Ω_0. By the Moreover clause of Lemma 6.2, the set

$$O^+ = \Omega_0 \setminus \bigcup_i \alpha_i((0,1))$$

is path connected and contains every outer tip $\alpha_i(1)$. Join the finitely many tips pairwise by paths in O^+; since $O^+ \setminus \{\text{tips}\} = \Omega_0 \setminus \bigcup_i \alpha_i((0,1])$ is open, the path interiors may be perturbed within that open set into general position, avoiding the finitely many tips except at their own endpoints, and then subdivided at their finitely many crossings. An inclusion-minimal connected subgraph G spanning the tip vertex set is a finite embedded tree with $G \cap \alpha_i = \{\alpha_i(1)\}$ and $G \cap \bigcup_i \alpha_i((0,1)) = \varnothing$. Attaching each port arc α_i to G at the single point $\alpha_i(1)$—the ports being pairwise disjoint and meeting G nowhere else—yields a finite embedded plane tree $T_{\mathfrak{R}} = G \cup \bigcup_i \alpha_i$, so (6.3) holds and the data form a connector datum. □

Lemma 6.7 (Proper protected impressions leave a port)**.** *Let* $K \Subset \operatorname{int} D$ *be a nondegenerate full continuum in a closed Jordan disk* D *and let* $P_{\text{prot}} \subsetneq \partial K$ *be compact, possibly empty. There are* $x \in \partial K \setminus P_{\text{prot}}$ *and a Jordan arc* α *from* x *to* ∂D*, otherwise in* $\operatorname{int} D \setminus K$*. Moreover, finitely many crosscut chains of* $\widehat{\mathbb{C}} \setminus K$ *whose impressions have union contained in* P_{prot} *can*

be truncated so that, writing S for the connected union of K with the truncated relative collar tails,

$$S \Subset \operatorname{int} D, \qquad \alpha \cap \operatorname{Fill}(S) = \{x\}. \tag{6.6}$$

Proof. If $P_{\mathrm{prot}} = \varnothing$, choose any $y \in \partial K$ and skip the distance argument below (there are then no chains to truncate, $S = K$, and only the port construction is needed). Otherwise choose $y \in \partial K \setminus P_{\mathrm{prot}}$ and put $\delta = \operatorname{dist}(y, P_{\mathrm{prot}}) > 0$. Choose $z \in \operatorname{int} D \setminus K$ with $|z - y| < \delta/4$, and let $x \in K$ minimize distance to z. The segment $(x, z]$ misses K, and $|x - y| \le 2|z - y| < \delta/2$, so $x \notin P_{\mathrm{prot}}$. The complement of the full continuum K in $\widehat{\mathbb{C}}$ is an open connected, hence arcwise connected, set; join z to the exterior of D, take a simple subarc of the concatenation starting at x, and stop at its first hit on ∂D. This is α.

The compact arc α is disjoint from P_{prot} and, when that set is nonempty, lies at positive distance from it. For each chain, choose an open neighborhood N of its impression with $\overline{N} \Subset \operatorname{int} D$ and $\overline{N} \cap \alpha = \varnothing$. The nested compact sets $\overline{D_n}$ have intersection equal to the impression, so compactness gives $\overline{D_n} \subset N$ for all sufficiently large n. Truncate each chain at such a depth. Because there are only finitely many chains and $K \Subset \operatorname{int} D$, the resulting connected compact set S satisfies $S \Subset \operatorname{int} D$ and $\alpha \cap S = \{x\}$.

Extend α from its endpoint on ∂D to ∞ along a Jordan arc otherwise in $\widehat{\mathbb{C}} \setminus D$. The extension misses S because $S \Subset \operatorname{int} D$. Thus every point of α except x lies in the unbounded component of $\mathbb{C} \setminus S$. Filling the bounded complementary components cannot add a point of $\alpha \setminus \{x\}$, which proves (6.6). □

Theorem 6.8 (Finite direct-decoration protected carrier)**.** *Fix a nonempty finite set $F \subset \mathtt{A}^{<\mathbb{N}}$ of addresses and finitely many (possibly zero) crosscut chains of $\widehat{\mathbb{C}} \setminus K_\epsilon$ (the* protected root chains*) whose relative terminal collar tails are pairwise disjoint and whose total impression is a proper compact subset $P_{\mathrm{prot}} \subsetneq \partial K_\epsilon$. Let U be a Jordan domain with $P \Subset U$. Then the chains can be truncated and transported by the inverse branches ϕ_w ($w \in F$), and there exist a port point $x_w \in \partial K_w$ and a port arc α_w for each $w \in F$, such that:*

- (i) *the filled transported sets $L_w = \operatorname{Fill}\big(K_w \cup (\text{transported relative collar tails})\big)$ satisfy $K_w \subset L_w \Subset \operatorname{int} D_w$, so they lie in the pairwise disjoint gap cells of Theorem 5.4;*
- (ii) *there is a compact connected full carrier $H \Subset U$ containing every selected fiber K_w ($w \in F$) with singleton attachment trace $\overline{H \setminus K_w} \cap K_w = \{x_w\}$, and meeting the compactum in exactly the selected fibers:*

$$H \cap \mathcal{H}_\partial = \bigcup_{w \in F} K_w; \tag{6.7}$$

- (iii) *every transported protected chain survives as a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, and the survival map is injective and preserves the cyclic order of the transported family in $\widehat{\mathbb{C}} \setminus H$;*
- (iv) *no local connectivity of any fiber is assumed or concluded.*

Proof. Step 1: root port and truncation. Apply Lemma 6.7 with $K = K_\epsilon$, the cell D of (5.3), and P_{prot}. It gives a point $x \in \partial K_\epsilon \setminus P_{\mathrm{prot}}$, a port arc α, and truncations for which (6.6) holds. The truncated relative tails are pairwise disjoint: by (6.1), each lies in its original terminal tail $\overline{D_1} \setminus K_\epsilon$, and those tails are disjoint by hypothesis.

Step 2: transport. For $w \in F$ put

$$S_w = \phi_w(S), \qquad L_w = \operatorname{Fill}(S_w), \qquad x_w = \phi_w(x), \qquad \alpha_w = \phi_w(\alpha).$$

Each ϕ_w restricts to an orientation-preserving homeomorphism of closed Jordan disks $D \to D_w$ ((H2), Theorem 5.4). Since $S \Subset \operatorname{int} D$ is compact and connected, Theorem 5.4(5.6), applied to

the root cell, first gives

$$\mathrm{Fill}(S) \Subset \operatorname{int} D.$$

Let $\Phi_w \colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ be the pointed Schoenflies extension of $\phi_w|_D \colon D \to D_w$ from Lemma 3.3. Its Fill identity now applies with the required containment and yields

$$L_w = \mathrm{Fill}(\phi_w(S)) = \Phi_w(\mathrm{Fill}(S)) = \phi_w(\mathrm{Fill}(S)) \Subset \operatorname{int} D_w.$$

Therefore

$$\alpha_w \cap L_w = \phi_w\big(\alpha \cap \mathrm{Fill}(S)\big) = \{x_w\}. \tag{6.8}$$

The global homeomorphism Φ_w carries the truncated crosscut chains of $\widehat{\mathbb{C}} \setminus K_\epsilon$ to crosscut chains of $\widehat{\mathbb{C}} \setminus K_w$, preserving selected sides, the null-chain condition, and cyclic order; on the truncated tails it agrees with ϕ_w. The displayed compact containment proves (i), and (6.8) says that α_w runs from x_w to ∂D_w, otherwise in $\operatorname{int} D_w \setminus L_w$.

Step 3: one free component. Each L_w is full, so Theorem 5.5(ii) makes $U \setminus \widetilde{\mathcal{H}}$ connected. For the connector datum, take atoms $A_w := K_w$, the transported collar tails, and the compact forbidden set $Z := \mathcal{H}_\partial \setminus \bigcup_{w \in F} K_w$; compactness follows because the selected fibers are clopen in $\mathcal{H}_\partial$. The set Q of (6.5) lies in $\widetilde{\mathcal{H}}$, hence

$$U \setminus \widetilde{\mathcal{H}} \;\subset\; \Omega_{\mathrm{free}} = U \setminus Q,$$

and the connected set $U \setminus \widetilde{\mathcal{H}}$ lies in one component Ω_0 of Ω_{free}.

The port interiors lie in that component: by (6.8) and (5.5),

$$\alpha_w((0,1]) \;\subset\; \big(\operatorname{int} D_w \setminus L_w\big) \cup \partial D_w \;\subset\; U \setminus \widetilde{\mathcal{H}},$$

because the modified-fiber separation established in the proof of Theorem 5.5(i), together with (5.4)–(5.5), gives $D_w \cap \widetilde{\mathcal{H}} = L_w$, and ∂D_w misses $\widetilde{\mathcal{H}}$. Also $\alpha_w \cap K_w \subset \alpha_w \cap L_w = \{x_w\}$ with $x_w \in \partial K_w$ free: $x = \phi_w^{-1}(x_w)$ is not in P_{prot} and hence x_w is not an impression of a transported chain. The arcs α_w are pairwise disjoint, lying in the pairwise disjoint closed cells D_w. Thus Ω_0 has simultaneous ports to every atom.

Step 4: connector and carrier. Theorem 6.6 yields a connector datum, and Theorem 6.5 realizes it as a compact connected full carrier $H = H_{\mathfrak{R}} \Subset U$ containing every K_w with singleton attachment traces $\{x_w\}$, missing Z, with every transported chain surviving as a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, injectively and in the cyclic order of the transported family. For the exact trace (6.7): $H \supset \bigcup_{w \in F} K_w$ by construction, and $H \setminus \bigcup_{w \in F} K_w = T_{\mathfrak{R}} \setminus \{x_w : w \in F\}$ lies in $\Omega_{\mathrm{free}} \cup \{\alpha_w(1)\}$, which misses $Z \cup \bigcup_{w \in F} K_w = \mathcal{H}_\partial$ except at the port points $x_w \in K_w$ already counted; hence $H \cap \mathcal{H}_\partial$ is exactly the union of the selected fibers. This proves (ii) and (iii). The arguments use crosscut chains and plane topology only, never a boundary parametrization of any fiber, proving (iv). □

7. Limitations and an open direction

The end fibers E_ω are nested puzzle intersections; they have not been identified with small Julia sets, and no renormalization structure on them is claimed. The covering in Theorem 4.5 maps $E_{a\omega}$ to E_ω, but it does not provide a first-return or self-return map on an individual end fiber.

The replacement theorem permits infinitely many finite addresses, while Theorem 6.8 concerns finitely many chains at finitely many finite fibers. No uniform incidence bound or second satellite portrait is established. A natural open direction is to ask whether the construction can be iterated through disconnected satellite stages while retaining a compact component quotient and a uniformly controlled access system.

These results were extracted from a larger research manuscript and made self-contained here. Repository notes retain the source concordance and the full list of boundary-case checks.

Machine-assistance disclosure. Drafting, source comparison, verification, and build tooling used Fae Build, Anthropic Fable 5, and OpenAI GPT-5.6 Sol. No model output is treated as mathematical authority; all mathematical claims and proofs are the author's responsibility.

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